

Gary Gallopin, week 1:

The following URL is for the cpp file strcmp_case_insensitive:

<https://github.com/ggallopin-dot/CaseInsensitive.git>

Also submit your responses to the following Thought Provoking Questions:

How might you compare two characters without reference to case? How might you do this without destroying the character variable(s) contents?

I used `tolower` that requires `#include <cctype>`. The function `tolower` turns Capital letters into lower case letters to make the comparisons case insensitive. I created a function called `compareStrings` which took the two strings to compare as arguments. I demonstrate that the strings being passed as arguments are not destroyed as all the manipulation of them is done within the function as parameters that are examined one character at a time and only there the function `tolower` is used.

How can you compare two strings in a case-insensitive way without destroying their contents?

As I mention above, by doing the case lowering inside a function. Inside the function itself, another approach could be using the ASCII numbers of each character, manipulating the upper case numbers to represent the lower case numbers of the corresponding letters thus never changing the characters themselves.

What kind of arguments should your string comparison function take? Are you going to pass these arguments by value or by reference?

My function, `compareStrings`, takes the value of the two strings to compare as two arguments.

Why do you think the return value of `strcmp` is "greater than zero", "less than zero", and "equal to zero" rather than -1, 0, 1 specifically?

Maybe to alert a programmer different than the creator as to what the function does.

What cases might you need to consider to test your function thoroughly? How many tests have you had to write to feel confident in your code?

I included the following kinds of cases: upper case vs. lower case; strings of different lengths; strings that included numbers; strings that included blanks or punctuation (I had to check the specific ASCII numbers for these to do the proper assert test.); strings that were a single character; and empty strings. I wrote a total of 15 assert tests.